LAB RECORD

1. a) Install Flutter and Dart SDK.

b) Write a simple Dart program to understand the language basics.

Ans:

Pre-Installed

Git ------

1.Open a Web browser

2.visit the official Git website

3. Download the git windows suitable for the operating system.

4.After Download open to setup the git in your system just clicking the next upto to finish.

https://git-scm.com/install/windows

Step 1: Download Flutter SDK

1. Open a web browser.
2. Visit the official Flutter website.
3. Download the Flutter SDK suitable for the operating system.
4. Extract the downloaded Flutter SDK to a suitable location, for example:

Steps:

```
# 1. Download the Flutter SDK
https://flutter.dev/docs/get-started/install

# 2. Extract the zip file to your desired location (e.g., C:\src\flutter).

# 3. Add Flutter to your PATH
# For Windows:
set PATH=%PATH%;C:\src\flutter\bin

# For macOS/Linux:
export PATH="$PATH:/path-to-flutter/bin"
```

The Flutter SDK provides tools and libraries to develop Flutter apps. Make sure the `flutter` command works in your terminal by adding it to the PATH.

Step 2: Set the Environment Variable

Add the Flutter bin directory to the system **PATH**.

C:\src\flutter\bin

Step 3: Verify Flutter Installation

Open **Command Prompt** or **Terminal** and execute:

flutter –version

Step 4: Run Flutter Doctor

Execute the following command:

flutter doctor

Step 5: Install Android Studio

1. Download and install Android Studio.
2. Install the Android SDK and Android SDK Platform Tools.
3. Configure an Android emulator if required.
4. Open Android Studio and install the Flutter and Dart plugins.

3. Set Up Android Studio

Android Studio is required for Flutter development to set up the Android emulator and required tools.

Steps:

```
# 1. Install Android Studio
https://developer.android.com/studio

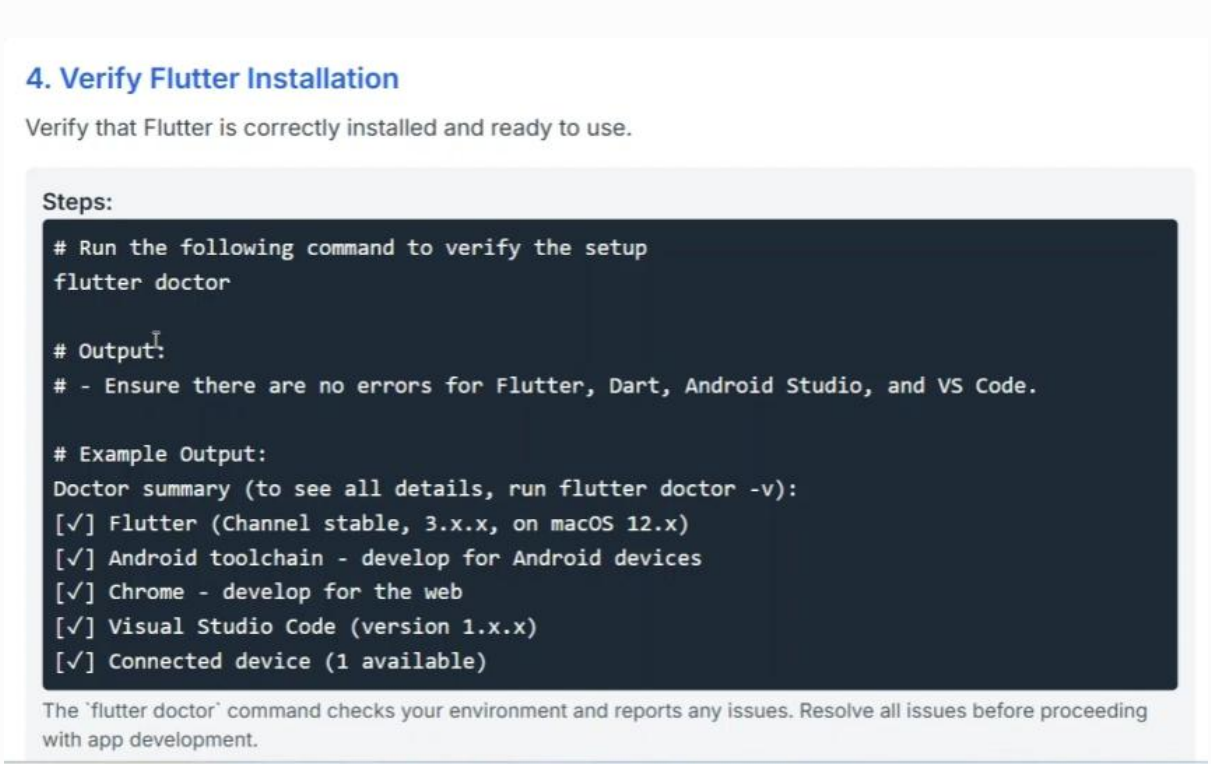
# 2. Install SDK Tools:
# - Open Android Studio
# - Go to Preferences/Settings > Appearance & Behavior > System Settings > Android SDK.
# - Install the 'Android SDK Platform-Tools' and the latest 'Android API'.

# 3. Set Up an Emulator:
# - Go to Tools > AVD Manager.
# - Create a new virtual device with your desired specifications.
```

Android Studio provides an emulator to test your Flutter applications on a virtual Android device.

Step 6: Verify the Development Environment

flutter doctor



installations are completed , now you can open the vs code

Go to the extensions

1.Download Flutter

2.Downlaod Dart

b) Write a simple Dart program to understand the language basics.

1.Open a web browser

2. Visit the Dev pad site to **Run** the sample **dart** programs.

Program:

```
void main() {
  // Variables
  String name = "Ravi";
  int age = 20;
  double marks = 85.5;
  bool passed = true;
```

```
  // Display output
  print("Student Details");
  print("----------------");
  print("Name: $name");
  print("Age: $age");
  print("Marks: $marks");
  print("Passed: $passed");

  // Arithmetic operation
  int nextYearAge = age + 1;
  print("Age next year: $nextYearAge");

  // Conditional statement
  if (marks >= 40) {
    print("Result: PASS");
  } else {
    print("Result: FAIL");
  }
}
```

Now Open Vs Code

1. To create a project

2.Before that you can Create a newfolder in your system(eg:Projects)

3.In Vs Code top search bar just give the greater than symbol(>flutter).

4.You can Application and Select the project folder and name the flutter_application_1 to app.

5.After few seconds ,then can generate the file structure from left side now you open **lib** folder , you can see sample code in main.dart or u can add the newfiles in lib folder only.

6.To Run **the program in Terminal just give the command**

flutter run

2. a) Explore various Flutter widgets (Text, Image, Container, etc.).

Ans:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}


class MyApp extends StatelessWidget {
  const MyApp({super.key});


  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
```

```
appBar: AppBar(
  title: const Text("Flutter Widgets"),
),

body: SingleChildScrollView(
  padding: const EdgeInsets.all(20),

  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [

      // 1. Text Widget
      const Text(
        "Welcome to Flutter",
        style: TextStyle(
          fontSize: 25,
          fontWeight: FontWeight.bold,
        ),
      ),

      const SizedBox(height: 20),

      // 2. Container Widget
      Container(
        width: double.infinity,
```

```
  padding: const EdgeInsets.all(20),
  decoration: BoxDecoration(
    color: Colors.blue,
    borderRadius: BorderRadius.circular(10),
  ),
  child: const Text(
    "This is a Container",
    style: TextStyle(
      color: Colors.white,
      fontSize: 18,
    ),
  ),
),

const SizedBox(height: 20),

// 3. Image Widget
Image.network(
  "https://picsum.photos/400/200",
  height: 200,
  width: double.infinity,
  fit: BoxFit.cover,
),

const SizedBox(height: 20),
```

```
// 4. Icon Widget
const Icon(
  Icons.favorite,
  color: Colors.red,
  size: 50,
),

const SizedBox(height: 20),

// 5. Card Widget
const Card(
  elevation: 5,
  child: Padding(
    padding: EdgeInsets.all(20),
    child: Text(
      "This is a Card Widget",
      style: TextStyle(fontSize: 18),
    ),
  ),
),

const SizedBox(height: 20),

// 6. Button Widget
```

```
            ElevatedButton(
              onPressed: () {
                print("Button Clicked");
              },
              child: const Text("Click Me"),
            ),
          ],
        ),
      ),
    ),
  );
 }
}
```

b) Implement different layout structures using Row, Column, and Stack widgets.

```
import 'package:flutter/material.dart';
void main() {
 runApp(const MyApp());
}

class MyApp extends StatelessWidget {
 const MyApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
```

```
debugShowCheckedModeBanner: false,
home: Scaffold(
  appBar: AppBar(
    title: const Text("Flutter Layouts"),
    backgroundColor: Colors.blue,
  ),
  body: SingleChildScrollView(
    child: Padding(
      padding: const EdgeInsets.all(20),
      child: Column(
        children: [

          // ROW LAYOUT
          const Text(
            "Row Layout",
            style: TextStyle(fontSize: 24),
          ),

          const SizedBox(height: 15),

          Row(
            mainAxisAlignment:
                MainAxisAlignment.spaceEvenly,
            children: [
              Container(
```

```
      width: 80,
      height: 80,
      color: Colors.red,
    ),
    Container(
      width: 80,
      height: 80,
      color: Colors.green,
    ),
    Container(
      width: 80,
      height: 80,
      color: Colors.blue,
    ),
  ],
),

const SizedBox(height: 30),

// COLUMN LAYOUT
const Text(
  "Column Layout",
  style: TextStyle(fontSize: 24),
),
```

```
const SizedBox(height: 15),

Column(
  children: [
    Container(
      width: 100,
      height: 60,
      color: Colors.orange,
    ),
    const SizedBox(height: 10),
    Container(
      width: 100,
      height: 60,
      color: Colors.purple,
    ),
    const SizedBox(height: 10),
    Container(
      width: 100,
      height: 60,
      color: Colors.teal,
    ),
  ],
),

const SizedBox(height: 30),
```

```
// STACK LAYOUT
const Text(
  "Stack Layout",
  style: TextStyle(fontSize: 24),
),

const SizedBox(height: 15),

SizedBox(
  width: 250,
  height: 200,
  child: Stack(
    children: [
      Container(
        width: 250,
        height: 200,
        color: Colors.yellow,
      ),
      Positioned(
        top: 30,
        left: 40,
        child: Container(
          width: 170,
          height: 130,
```

```
                color: Colors.blue,
              ),
            ),
            Positioned(
              top: 65,
              left: 80,
              child: Container(
                width: 90,
                height: 60,
                color: Colors.red,
              ),
            ),
          ],
        ),
      ),
    ),
  ),
 );
 }
}
```

3. a) Design a responsive UI that adapts to different screen sizes.

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {

    // Get screen width
```

```
double width = MediaQuery.of(context).size.width;

return Scaffold(
  appBar: AppBar(
    title: const Text("Responsive UI"),
  ),

  body: Center(
    child: width < 600
        ? Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              box("Home"),
              box("Profile"),
              box("Settings"),
            ],
          )
        : Row(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              box("Home"),
              box("Profile"),
              box("Settings"),
            ],
          ),
```

```
      ),
    );
  }

  // Reusable widget
  Widget box(String title) {
    return Container(
      width: 120,
      height: 80,
      margin: const EdgeInsets.all(10),
      alignment: Alignment.center,
      color: Colors.blue,
      child: Text(
        title,
        style: const TextStyle(
          color: Colors.white,
          fontSize: 18,
        ),
      ),
    );
  }
}
```

Errors u can get to modify the lengths

Mobile → 390 pixels

Tablet → 800 pixels

Desktop → 1200 pixels

b) Implement media queries and breakpoints for responsiveness.

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  // Create a reusable box
  Widget itemBox(String name) {
    return Container(
```

```
    height: 100,
    margin: const EdgeInsets.all(8),
    alignment: Alignment.center,
    decoration: BoxDecoration(
      color: Colors.blue,
      borderRadius: BorderRadius.circular(10),
    ),
    child: Text(
      name,
      style: const TextStyle(
        color: Colors.white,
        fontSize: 18,
      ),
    ),
  );
}

@override
Widget build(BuildContext context) {

  // Get screen width
  double width = MediaQuery.of(context).size.width;

  int columns;
```

```
// Breakpoints
if (width < 600) {
  columns = 1;       // Mobile
} else if (width < 1000) {
  columns = 2;       // Tablet
} else {
  columns = 3;       // Desktop
}

return Scaffold(
  appBar: AppBar(
    title: const Text("Media Query Demo"),
  ),

  body: Padding(
    padding: const EdgeInsets.all(10),

    child: Column(
      children: [

        Text(
          "Screen Width: ${width.toStringAsFixed(0)}",
          style: const TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
```

```
            ),
          ),

          const SizedBox(height: 20),

          Expanded(
            child: GridView.count(
              crossAxisCount: columns,
              children: [
                itemBox("Home"),
                itemBox("Profile"),
                itemBox("Courses"),
                itemBox("Students"),
                itemBox("Faculty"),
                itemBox("Settings"),
              ],
            ),
          ),
        ],
      ),
    ),
  );
 }
}
```

4. a) Set up navigation between different screens using Navigator.

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomeScreen(),
    );
  }
}

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
```

```
      title: const Text("Home Screen"),
    ),

    body: Center(
      child: ElevatedButton(
        child: const Text("Go to Second Screen"),

        onPressed: () {

          // Navigate to Second Screen
          Navigator.push(
            context,
            MaterialPageRoute(
              builder: (context) => const SecondScreen(),
            ),
          );
        },
      ),
    ),
  );
 }
}

// ---------- SECOND SCREEN ----------
```

```
class SecondScreen extends StatelessWidget {
  const SecondScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Second Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go Back"),

          onPressed: () {

            // Return to previous screen
            Navigator.pop(context);
          },
        ),
      ),
    );
  }
}
```

b) Implement navigation with named routes.

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,

      // Define named routes
      initialRoute: '/',

      routes: {
        '/': (context) => const HomeScreen(),
        '/profile': (context) => const ProfileScreen(),
        '/settings': (context) => const SettingsScreen(),
      },
    );
  }
```

```
}

// ---------- HOME SCREEN ----------

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Home Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go to Profile"),

          onPressed: () {
            Navigator.pushNamed(
              context,
              '/profile',
            );
          },
        ),
```

```
      ),
    );
  }
}

// ---------- PROFILE SCREEN ----------

class ProfileScreen extends StatelessWidget {
  const ProfileScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Profile Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go to Settings"),

          onPressed: () {
            Navigator.pushNamed(
              context,
              '/settings',
```

```
            );
          },
        ),
      ),
    );
  }
}

// ---------- SETTINGS SCREEN ----------

class SettingsScreen extends StatelessWidget {
  const SettingsScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Settings Screen"),
      ),

      body: Center(
        child: ElevatedButton(
          child: const Text("Go Back"),

          onPressed: () {
```

```
          Navigator.pop(context);
        },
      ),
    ),
  );
 }
}
```

5. a) Learn about stateful and stateless widgets.

What is a StatelessWidget?

A StatelessWidget is a widget whose data does not change while the application is running.

```
class Welcome extends StatelessWidget {
  const Welcome({super.key});
  @override
  Widget build(BuildContext context) {
    return const Text("Welcome to Flutter");
  }
}
```

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

// ---------- STATELESS WIDGET ----------
```

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const CounterPage(),
    );
  }
}
// ---------------- STATEFUL WIDGET ----------------
class CounterPage extends StatefulWidget {
  const CounterPage({super.key});
  @override
  State<CounterPage> createState() => _CounterPageState();
}
// State of CounterPage
class _CounterPageState extends State<CounterPage> {

  int count = 0;

  // Increase counter
  void increase() {
    setState(() {
```

```
    count++;
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("Stateful Widget"),
    ),

    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          // Stateless Text
          const Text(
            "Counter Value",
            style: TextStyle(
              fontSize: 25,
              fontWeight: FontWeight.bold,
            ),
          ),
```

```
            const SizedBox(height: 20),

            // Changing value
            Text(
              "$count",
              style: const TextStyle(
                fontSize: 50,
                color: Colors.blue,
              ),
            ),

            const SizedBox(height: 20),

            ElevatedButton(
              onPressed: increase,
              child: const Text("Increase"),
            ),
          ],
        ),
      ),
    );
  }
}
```

b) Implement state management using set State and Provider.

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (context) => CounterProvider(),
      child: const MyApp(),
    ),
  );
}

// ---------------- PROVIDER ----------------

class CounterProvider extends ChangeNotifier {

  int count = 0;

  void increase() {
    count++;
    notifyListeners();
  }

  void decrease() {
```

```
    count--;
    notifyListeners();
  }
}

// ---------------- APP ----------------

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const CounterPage(),
    );
  }
}

// ---------------- UI ----------------

class CounterPage extends StatelessWidget {
  const CounterPage({super.key});

  @override
```

```
Widget build(BuildContext context) {

 final counter = Provider.of<CounterProvider>(context);

 return Scaffold(
  appBar: AppBar(
   title: const Text("Provider Example"),
  ),

  body: Center(
   child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [

     const Text(
      "Counter",
      style: TextStyle(
       fontSize: 25,
      ),
     ),

     const SizedBox(height: 20),

     Text(
      "${counter.count}",
```

```
  style: const TextStyle(
    fontSize: 50,
    fontWeight: FontWeight.bold,
  ),
),

const SizedBox(height: 20),

Row(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [

    ElevatedButton(
      onPressed: counter.decrease,
      child: const Text("-"),
    ),

    const SizedBox(width: 20),

    ElevatedButton(
      onPressed: counter.increase,
      child: const Text("+"),
    ),
  ],
),
```

```
          ],
        ),
      ),
    );
  }
}
```

6. a) Create custom widgets for specific UI elements.

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}
```

```
// ---------------- HOME PAGE ----------------

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Custom Widgets"),
      ),

      body: Padding(
        padding: const EdgeInsets.all(15),

        child: Column(
          children: [

            StudentCard(
              name: "Ravi",
              course: "Data Science",
            ),

            StudentCard(
              name: "Sita",
```

```
            course: "Computer Science",
          ),

          StudentCard(
            name: "Rahul",
            course: "Artificial Intelligence",
          ),
        ],
      ),
    ),
  );
 }
}

// ---------------- CUSTOM WIDGET ----------------

class StudentCard extends StatelessWidget {

 final String name;
 final String course;

 const StudentCard({
  super.key,
  required this.name,
  required this.course,
```

```
  });

  @override
  Widget build(BuildContext context) {
    return Card(
      elevation: 4,
      margin: const EdgeInsets.only(bottom: 15),

      child: ListTile(
        leading: const CircleAvatar(
          child: Icon(Icons.person),
        ),

        title: Text(
          name,
          style: const TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ),
        ),
        subtitle: Text(course),
      ),
    );
  }
}
```

b) Apply styling using themes and custom styles.

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,

      // Application Theme
      theme: ThemeData(
        primarySwatch: Colors.blue,

        appBarTheme: const AppBarTheme(
          backgroundColor: Colors.blue,
          foregroundColor: Colors.white,
        ),

        elevatedButtonTheme: ElevatedButtonThemeData(
```

```
          style: ElevatedButton.styleFrom(
            backgroundColor: Colors.blue,
            foregroundColor: Colors.white,
          ),
        ),
      ),

      home: const HomePage(),
    );
  }
}

// ---------- HOME PAGE ----------

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {

    // Custom Text Styles
    const titleStyle = TextStyle(
      fontSize: 25,
      fontWeight: FontWeight.bold,
      color: Colors.blue,
```

```
);

const normalStyle = TextStyle(
  fontSize: 18,
  color: Colors.black87,
);

return Scaffold(
  appBar: AppBar(
    title: const Text("Theme & Styles"),
  ),

  body: Padding(
    padding: const EdgeInsets.all(20),

    child: Column(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [

        // Custom style
        const Text(
          "Student Profile",
          style: titleStyle,
        ),
```

```
const SizedBox(height: 20),

const Text(
  "Name: Ravi",
  style: normalStyle,
),

const SizedBox(height: 10),

const Text(
  "Course: Data Science",
  style: normalStyle,
),

const SizedBox(height: 20),

// Card with custom decoration
Card(
  elevation: 5,
  shape: RoundedRectangleBorder(
    borderRadius: BorderRadius.circular(15),
  ),

  child: const Padding(
    padding: EdgeInsets.all(20),
```

```
    child: Row(
      children: [
        CircleAvatar(
          radius: 30,
          child: Icon(Icons.person),
        ),

        SizedBox(width: 20),

        Text(
          "Welcome, Ravi!",
          style: TextStyle(
            fontSize: 20,
            fontWeight: FontWeight.bold,
          ),
        ),
      ],
    ),
  ),
),

const SizedBox(height: 30),

// Theme applied to button
```

```
            ElevatedButton(
              onPressed: () {},
              child: const Text("Continue"),
            ),
          ],
        ),
      ),
    );
  }
}
```

7. a) Design a form with various input fields.

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const RegistrationPage(),
    );
```

```
  }
}

// ---------------- FORM PAGE ----------------

class RegistrationPage extends StatefulWidget {
  const RegistrationPage({super.key});

  @override
  State<RegistrationPage> createState() =>
      _RegistrationPageState();
}

class _RegistrationPageState extends State<RegistrationPage> {

  // Controllers
  final nameController = TextEditingController();
  final emailController = TextEditingController();
  final phoneController = TextEditingController();
  final passwordController = TextEditingController();

  String course = "Data Science";

  @override
  Widget build(BuildContext context) {
```

```
return Scaffold(
  appBar: AppBar(
    title: const Text("Student Registration"),
  ),

  body: Padding(
    padding: const EdgeInsets.all(20),

    child: Form(
      child: Column(
        children: [

          // Name
          TextFormField(
            controller: nameController,
            decoration: const InputDecoration(
              labelText: "Name",
              hintText: "Enter your name",
              prefixIcon: Icon(Icons.person),
              border: OutlineInputBorder(),
            ),
          ),

          const SizedBox(height: 15),
```

```
// Email
TextFormField(
  controller: emailController,
  keyboardType: TextInputType.emailAddress,
  decoration: const InputDecoration(
    labelText: "Email",
    hintText: "Enter your email",
    prefixIcon: Icon(Icons.email),
    border: OutlineInputBorder(),
  ),
),

const SizedBox(height: 15),

// Phone
TextFormField(
  controller: phoneController,
  keyboardType: TextInputType.phone,
  decoration: const InputDecoration(
    labelText: "Phone Number",
    hintText: "Enter phone number",
    prefixIcon: Icon(Icons.phone),
    border: OutlineInputBorder(),
  ),
),
```

```
const SizedBox(height: 15),

// Password
TextFormField(
  controller: passwordController,
  obscureText: true,
  decoration: const InputDecoration(
    labelText: "Password",
    hintText: "Enter password",
    prefixIcon: Icon(Icons.lock),
    border: OutlineInputBorder(),
  ),
),

const SizedBox(height: 15),

// Course Dropdown
DropdownButtonFormField<String>(
  value: course,
  decoration: const InputDecoration(
    labelText: "Course",
    border: OutlineInputBorder(),
  ),
  items: const [
```

```
    DropdownMenuItem(
      value: "Data Science",
      child: Text("Data Science"),
    ),
    DropdownMenuItem(
      value: "Computer Science",
      child: Text("Computer Science"),
    ),
    DropdownMenuItem(
      value: "Artificial Intelligence",
      child: Text("Artificial Intelligence"),
    ),
  ],
  onChanged: (value) {
    setState(() {
      course = value!;
    });
  },
),

const SizedBox(height: 25),

// Submit Button
SizedBox(
  width: double.infinity,
```

```
              child: ElevatedButton(
                onPressed: () {
                  print("Name: ${nameController.text}");
                  print("Email: ${emailController.text}");
                  print("Phone: ${phoneController.text}");
                  print("Course: $course");
                },
                child: const Text("Register"),
              ),
            ),
          ],
        ),
      ),
    ),
  );
 }
}
```

b) Implement form validation and error handling.

```
import 'package:flutter/material.dart';

void main() {
 runApp(const MyApp());
}

class MyApp extends StatelessWidget {
```

```
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const RegistrationPage(),
    );
  }
}

// ---------- REGISTRATION PAGE ----------

class RegistrationPage extends StatefulWidget {
  const RegistrationPage({super.key});

  @override
  State<RegistrationPage> createState() =>
      _RegistrationPageState();
}

class _RegistrationPageState extends State<RegistrationPage> {

  // Form key
  final formKey = GlobalKey<FormState>();
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("Form Validation"),
    ),

    body: Padding(
      padding: const EdgeInsets.all(20),

      child: Form(
        key: formKey,

        child: Column(
          children: [

            // Name
            TextFormField(
              decoration: const InputDecoration(
                labelText: "Name",
                border: OutlineInputBorder(),
              ),

              validator: (value) {
```

```
    if (value == null || value.isEmpty) {
      return "Please enter your name";
    }

    return null;
  },
),

const SizedBox(height: 15),

// Email
TextFormField(
  keyboardType: TextInputType.emailAddress,

  decoration: const InputDecoration(
    labelText: "Email",
    border: OutlineInputBorder(),
  ),

  validator: (value) {
    if (value == null || value.isEmpty) {
      return "Please enter your email";
    }

    if (!value.contains("@")) {
```

```
      return "Enter a valid email";
    }

    return null;
  },
),

const SizedBox(height: 15),

// Phone
TextFormField(
  keyboardType: TextInputType.phone,

  decoration: const InputDecoration(
    labelText: "Phone Number",
    border: OutlineInputBorder(),
  ),

  validator: (value) {
    if (value == null || value.isEmpty) {
      return "Please enter phone number";
    }

    if (value.length != 10) {
      return "Phone number must be 10 digits";
```

```
    }

    return null;
  },
),

const SizedBox(height: 15),

// Password
TextFormField(
  obscureText: true,

  decoration: const InputDecoration(
    labelText: "Password",
    border: OutlineInputBorder(),
  ),

  validator: (value) {
    if (value == null || value.isEmpty) {
      return "Please enter password";
    }

    if (value.length < 6) {
      return "Password must be at least 6 characters";
    }
```

```
    return null;
  },
),

const SizedBox(height: 25),

// Submit Button
SizedBox(
  width: double.infinity,

  child: ElevatedButton(
    onPressed: () {

      // Validate form
      if (formKey.currentState!.validate()) {

        ScaffoldMessenger.of(context).showSnackBar(
          const SnackBar(
            content: Text("Registration Successful"),
          ),
        );

      }
    },
```

```
                child: const Text("Register"),
              ),
            ),
          ],
        ),
      ),
    ),
  );
 }
}
```

8. a) Add animations to UI elements using Flutter's animation framework.

```
import 'package:flutter/material.dart';
void main() {
 runApp(const MyApp());
}
class MyApp extends StatelessWidget {
 const MyApp({super.key});
 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   debugShowCheckedModeBanner: false,
   home: const AnimationPage(),
  );
 }
```

```
}

class AnimationPage extends StatefulWidget {
  const AnimationPage({super.key});

  @override
  State<AnimationPage> createState() => _AnimationPageState();
}

class _AnimationPageState extends State<AnimationPage> {

  // Animation value
  double opacity = 0.0;

  void startAnimation() {
    setState(() {
      opacity = 1.0;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Fade Animation"),
```

```
),

body: Center(
  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [

      // Animated Widget
      AnimatedOpacity(
        opacity: opacity,
        duration: const Duration(seconds: 2),

        child: Container(
          width: 200,
          height: 200,
          color: Colors.blue,

          child: const Center(
            child: Text(
              "Hello Flutter",
              style: TextStyle(
                color: Colors.white,
                fontSize: 22,
              ),
            ),
```

```
            ),
          ),
        ),

        const SizedBox(height: 30),

        // Button
        ElevatedButton(
          onPressed: startAnimation,
          child: const Text("Start Animation"),
        ),
      ],
    ),
  ),
 );
 }
}
```

b) Experiment with different types of animations (fade, slide, etc.).

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const AnimationPage(),
    );
  }
}

class AnimationPage extends StatefulWidget {
  const AnimationPage({super.key});

  @override
  State<AnimationPage> createState() => _AnimationPageState();
}

class _AnimationPageState extends State<AnimationPage> {
```

```
bool animate = false;

void startAnimation() {
  setState(() {
    animate = !animate;
  });
}

@override
Widget build(BuildContext context) {

  return Scaffold(
    appBar: AppBar(
      title: const Text("Different Animations"),
    ),

    body: SingleChildScrollView(
      child: Center(
        child: Column(
          children: [

            const SizedBox(height: 30),

            // FADE ANIMATION
            const Text(
```

```
  "Fade Animation",
  style: TextStyle(
    fontSize: 20,
    fontWeight: FontWeight.bold,
  ),
),

const SizedBox(height: 10),

AnimatedOpacity(
  opacity: animate ? 0.2 : 1.0,
  duration: const Duration(seconds: 1),

  child: box("Fade"),
),

const SizedBox(height: 30),

// SLIDE ANIMATION
const Text(
  "Slide Animation",
  style: TextStyle(
    fontSize: 20,
    fontWeight: FontWeight.bold,
  ),
```

```
),

const SizedBox(height: 10),

AnimatedContainer(
  duration: const Duration(seconds: 1),

  transform: Matrix4.translationValues(
    animate ? 100 : 0,
    0,
    0,
  ),

  child: box("Slide"),
),

const SizedBox(height: 30),

// SCALE ANIMATION
const Text(
  "Scale Animation",
  style: TextStyle(
    fontSize: 20,
    fontWeight: FontWeight.bold,
  ),
```

```
),

const SizedBox(height: 10),

AnimatedScale(
  scale: animate ? 1.5 : 1.0,
  duration: const Duration(seconds: 1),

  child: box("Scale"),
),

const SizedBox(height: 50),

// ROTATION ANIMATION
const Text(
  "Rotation Animation",
  style: TextStyle(
    fontSize: 20,
    fontWeight: FontWeight.bold,
  ),
),

const SizedBox(height: 10),

AnimatedRotation(
```

```
            turns: animate ? 0.5 : 0,
            duration: const Duration(seconds: 1),

            child: box("Rotate"),
          ),

          const SizedBox(height: 40),

          ElevatedButton(
            onPressed: startAnimation,
            child: const Text("Start Animation"),
          ),
        ],
      ),
    ),
  );
}

// Reusable box
Widget box(String text) {
  return Container(
    width: 120,
    height: 80,
    alignment: Alignment.center,
```

```
      decoration: BoxDecoration(
        color: Colors.blue,
        borderRadius: BorderRadius.circular(10),
      ),

      child: Text(
        text,
        style: const TextStyle(
          color: Colors.white,
          fontSize: 18,
        ),
      ),
    );
  }
}
```

9. a) Fetch data from a REST API.

What is a REST API?

A REST API allows an application to communicate with a server and exchange data.

Add the HTTP Package

Open the VS Code terminal inside your Flutter project:

flutter pub add http

```
import 'dart:convert';


import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;


void main() {
  runApp(const MyApp());
}


class MyApp extends StatelessWidget {
  const MyApp({super.key});


  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const ApiPage(),
    );
  }
}
```

```
// ---------- API PAGE ----------

class ApiPage extends StatefulWidget {
  const ApiPage({super.key});

  @override
  State<ApiPage> createState() => _ApiPageState();
}

class _ApiPageState extends State<ApiPage> {

  // Fetch data from API
  Future<List<dynamic>> fetchPosts() async {

    final response = await http.get(
      Uri.parse(
        'https://jsonplaceholder.typicode.com/posts',
      ),
    );

    if (response.statusCode == 200) {
      return jsonDecode(response.body);
    } else {
      throw Exception("Failed to load data");
    }
  }
```

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("REST API Data"),
    ),

    body: FutureBuilder<List<dynamic>>(
      future: fetchPosts(),

      builder: (context, snapshot) {

        // Loading
        if (snapshot.connectionState ==
            ConnectionState.waiting) {
          return const Center(
            child: CircularProgressIndicator(),
          );
        }

        // Error
        if (snapshot.hasError) {
          return Center(
            child: Text(
              "Error: ${snapshot.error}",
```

```
    ),
  );
}

// Display data
final posts = snapshot.data!;

return ListView.builder(
  itemCount: posts.length,

  itemBuilder: (context, index) {

    return Card(
      margin: const EdgeInsets.all(10),

      child: ListTile(
        leading: CircleAvatar(
          child: Text(
            "${posts[index]['id']}",
          ),
        ),

        title: Text(
          posts[index]['title'],
          style: const TextStyle(
            fontWeight: FontWeight.bold,
```

```
                ),
              ),

              subtitle: Text(
                posts[index]['body'],
              ),
            ),
          );
        },
      );
    },
  ),
);
  }
}
```

b) Display the fetched data in a meaningful way in the UI.

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: "Student Posts",
      theme: ThemeData(
        primarySwatch: Colors.blue,
      ),
      home: const HomePage(),
    );
  }
}

class HomePage extends StatefulWidget {
  const HomePage({super.key});
```

```
  @override
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {

  Future<List<dynamic>> fetchPosts() async {
    final response = await http.get(
      Uri.parse(
        "https://jsonplaceholder.typicode.com/posts",
      ),
    );

    if (response.statusCode == 200) {
      return jsonDecode(response.body);
    } else {
      throw Exception("Failed to fetch data");
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Posts"),
```

```
  centerTitle: true,
),

body: FutureBuilder<List<dynamic>>(
  future: fetchPosts(),

  builder: (context, snapshot) {

    // Loading
    if (snapshot.connectionState == ConnectionState.waiting) {
      return const Center(
        child: CircularProgressIndicator(),
      );
    }

    // Error
    if (snapshot.hasError) {
      return Center(
        child: Text(
          "Error: ${snapshot.error}",
          style: const TextStyle(fontSize: 18),
        ),
      );
    }

    // Data
```

```
final posts = snapshot.data!;

return ListView.builder(
  padding: const EdgeInsets.all(10),
  itemCount: posts.length,

  itemBuilder: (context, index) {

    return Card(
      elevation: 4,
      margin: const EdgeInsets.only(bottom: 12),

      child: ListTile(

        leading: CircleAvatar(
          child: Text(
            "${posts[index]['id']}",
          ),
        ),

        title: Text(
          posts[index]['title'],
          style: const TextStyle(
            fontSize: 18,
            fontWeight: FontWeight.bold,
          ),
```

```
                ),

                subtitle: Padding(
                  padding: const EdgeInsets.only(top: 8),
                  child: Text(
                    posts[index]['body'],
                    style: const TextStyle(
                      fontSize: 15,
                    ),
                  ),
                ),
              ),
            ),
          );
        },
      );
    },
  );
 }
}
```

10. a) Write unit tests for UI components.
11.

```
app/
│
├── lib/
│   └── main.dart
│
├── test/
│   └── widget_test.dart
│
└── pubspec.yaml
```

Main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'UI Testing',
      home: const CounterPage(),
    );
  }
}

class CounterPage extends StatefulWidget {
  const CounterPage({super.key});

  @override
  State<CounterPage> createState() => _CounterPageState();
}

class _CounterPageState extends State<CounterPage> {
  int count = 0;

  // Increase counter
  void increase() {
```

```
  setState(() {
    count++;
  });
}

// Decrease counter
void decrease() {
  setState(() {
    count--;
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("UI Testing"),
      centerTitle: true,
    ),

    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          // Title
          const Text(
            "Counter",
            style: TextStyle(
              fontSize: 25,
              fontWeight: FontWeight.bold,
            ),
          ),

          const SizedBox(height: 20),

          // Counter value
          Text(
            "$count",
            style: const TextStyle(
              fontSize: 50,
              color: Colors.blue,
              fontWeight: FontWeight.bold,
```

```
            ),
          ),

          const SizedBox(height: 30),

          // Buttons
          Row(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [

              ElevatedButton(
                onPressed: decrease,
                child: const Text("Decrease"),
              ),

              const SizedBox(width: 20),

              ElevatedButton(
                onPressed: increase,
                child: const Text("Increase"),
              ),
            ],
          ),
        ],
      ),
    ),
   );
  }
}
```

test/widget_test.dart

```
import 'package:flutter_test/flutter_test.dart';

import 'package:ui_testing/main.dart';

void main() {

  testWidgets(
    'Counter UI test',
    (WidgetTester tester) async {

      // Load the application
      await tester.pumpWidget(const MyApp());
```

```
// Test 1: Check AppBar title
expect(
  find.text('UI Testing'),
  findsOneWidget,
);

// Test 2: Check Counter text
expect(
  find.text('Counter'),
  findsOneWidget,
);

// Test 3: Check initial counter value
expect(
  find.text('0'),
  findsOneWidget,
);

// Test 4: Check Increase button
expect(
  find.text('Increase'),
  findsOneWidget,
);

// Test 5: Check Decrease button
expect(
  find.text('Decrease'),
  findsOneWidget,
);

// Test 6: Tap Increase button
await tester.tap(
  find.text('Increase'),
);

// Rebuild the UI
await tester.pump();

// Counter should become 1
expect(
  find.text('1'),
  findsOneWidget,
```

```
    );

    // Test 7: Tap Increase again
    await tester.tap(
      find.text('Increase'),
    );

    await tester.pump();

    // Counter should become 2
    expect(
      find.text('2'),
      findsOneWidget,
    );

    // Test 8: Tap Decrease
    await tester.tap(
      find.text('Decrease'),
    );

    await tester.pump();

    // Counter should become 1
    expect(
      find.text('1'),
      findsOneWidget,
    );
  },
 );
}
```

b) Use Flutter's debugging tools to identify and fix issues.

Tools Used

- VS Code
- Flutter Debugger
- Debug Console
- Breakpoints
- print()
- debugPrint()
- Flutter DevTools
- Hot Reload

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const DebugPage(),
    );
  }
}
class DebugPage extends StatefulWidget {
  const DebugPage({super.key});
  @override
  State<DebugPage> createState() => _DebugPageState();
}
class _DebugPageState extends State<DebugPage> {
  int count = 0;
```

```
void increaseCounter() {
  setState(() {
    count++;
  });

  debugPrint("Counter value: $count");
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text("Flutter Debugging"),
    ),
    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          const Text(
            "Counter",
            style: TextStyle(
              fontSize: 25,
              fontWeight: FontWeight.bold,
            ),
          ),

          const SizedBox(height: 20),
```

```
            Text(
              "$count",
              style: const TextStyle(
                fontSize: 50,
                color: Colors.blue,
              ),
            ),

            const SizedBox(height: 20),

            ElevatedButton(
              onPressed: increaseCounter,
              child: const Text("Increase"),
            ),
          ],
        ),
      ),
    );
  }
}
```